Building an End-to-End Data Pipeline for NYC Airbnb Market Analytics

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***Abstract*— This study presents a comprehensive analysis of the New York City (NYC) Airbnb market using a dataset from 2019. A major challenge addressed in this work involves structural data anomalies, specifically text-delimiters causing widespread column misalignment where descriptive listing fields shifted into numeric ID columns. To resolve these data quality issues, an automated Extraction, Transformation, and Loading (ETL) pipeline was designed and executed using Pentaho Data Integration. The resulting cleansed data was then modeled and analyzed within an interactive Power BI dashboard environment. Visual insights reveal that Manhattan and Brooklyn heavily dominate the market, controlling a combined 85% of all active listings. Furthermore, renting out entire homes significantly boosts nightly revenue potential compared to private or shared spaces, while Brooklyn operates as the city's highest-velocity hub for guest turnover and transactional volume.**

Keywords*— *ETL, Pentaho, Power BI, Data Cleaning, Dashboard, Airbnb

I. Introduction

The sharing economy phenomenon, encouraged by collaborative consumption services such as Airbnb, has introduced a new paradigm in the city where residents' lodging is concerned and where the traditional hotel and hospitality industries are changing [1]. In cities such as New York City (NYC), millions of rental transactions take place every year, producing large amounts of behavioral, spatial and financial data. This data is invaluable for business analysts, real estate investors, and data professionals, providing a foundation to understand market trends, inform pricing strategies, and make informed commercial investments.

But it is not always easy to get any useful data from the raw public repositories (like CSV exports created by users). In reality, the data quality problems in real-world datasets can be significant and can negatively impact on the downstream analytics if not addressed [3]. One of the more common problems with tabular text is the embedding of unescaped text delimiters, such as raw commas within the description of the items listed. These characters move the columns in an unexpected way during parsing. As a result, textual data ends up in numeric-only columns, like unique identifier (ID). If this wrongly fed information is directly fed to the visualization tool it will break the relational schema, introduce mathematical biases and corrupt the final analytical results [4].

To overcome these practical engineering challenges, this paper presents a systematic end-to-end workflow for three key goals:

1) To resolve these data quality issues, an automated ETL pipeline was designed using Pentaho Data Integration to filter invalid rows, standardize data types, remove duplicates, and load clean records into an analytical database.

2) Build a strong, user-friendly and interactive data model and analytics dashboard in Power BI [5].

3) Glean strategic business knowledge about property supply trends, premium pricing trends and geographical occupancy trends throughout NYC.

II. Methodology

A. Dataset Characteristics

This study relies on the Airbnb listing data from NYC that was collected in 2019 uploaded on Kaggle by user “Dgomonov” [9]. Some of the raw data that's included are key business variables like neighborhood groups (boroughs), specific localized neighborhoods, spatial coordinates (latitude and longitude), room configurations, nightly rates and metrics of guest engagement (number of reviews, monthly review velocity).

B. Data Storage Design

The storage layer is divided into two main areas: raw storage and processed storage. The raw storage keeps the original Airbnb dataset as initially ingested from the source. This layer is important because it preserves the original data for auditing, troubleshooting, and reprocessing if errors are found.

The processed storage contains cleaned and structured tables that are ready for analytics. In this project, the processed data is stored in an OLAP MySQL database and later connected to Power BI. This separation ensures that raw data remains unchanged, while transformed data can be optimized for dashboard reporting.

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C. ETL Pipeline and Data Cleansing via Pentaho

Rather than executing complex string reconstruction on raw text, the engineering architecture takes advantage of a pre-staged relational database management system. The raw data was initially ingested into a MySQL database instance, where structural column shifts and format violations were filtered out or rejected by the database's strict schema constraints.

The Pentaho Data Integration (PDI) workflow was designed to act as a migration and feature-delivery pipeline. The operational pipeline consists of three sequential data-stream phases:

1) Database Source Ingestion: The workflow begins with a Table Input step connected directly to a local MySQL database via a JDBC driver, simulating an OLTP database. This project applies a full load strategy, where the complete Airbnb dataset is extracted from the source database and loaded into the target analytical database. Full load was selected because the dataset is static and represents NYC Airbnb listings from 2019, so incremental updates are not required.

2) Metadata Optimization and Field Mapping: A Select Values and Remove Duplicates steps are implemented immediately downstream from the source of ingestion. This step performs dual functions: it enforces strict schema consistency (e.g., casting geographic coordinates into high-precision floats and numerical attributes into integers) and renames or filters out redundant metadata to avoid any errors.

3) Target Serialization for Business Intelligence: The final stage routes the standardized data stream into a Table Output step. This component inserts the clean, relational rows into an OLAP MySQL database. This step inserts the clean and standardized relational records into the OLAP MySQL database. By storing the processed data in structured database tables with consistent field types, Power BI can consume the data reliably and build dashboard relationships without data parsing or modeling issues.

D. Data Quality and Validation

Several data quality checks were implemented to ensure that the processed dataset is reliable for analysis. First, duplicate listing records were removed using the Remove Duplicates step in Pentaho. Second, numeric fields such as price, minimum_nights, number_of_reviews, and availability_365 were validated to ensure that they contained valid numerical values. Third, geographic fields such as latitude and longitude were checked to ensure they were stored using the correct decimal format. Fourth, categorical fields such as neighbourhood_group and room_type were standardized to prevent inconsistent labels.

Records with severe structural issues, such as shifted columns or invalid ID fields, were filtered out or corrected before being loaded into the analytical database. These checks help prevent incorrect aggregations and misleading dashboard results.

E. Automation and Orchestration

The ETL workflow was automated using a Pentaho Job. The job controls the execution order of the pipeline, starting from source extraction, followed by data cleaning, duplicate removal, field mapping, and loading into the target OLAP database. This orchestration ensures that each transformation step runs in the correct sequence and reduces the need for manual execution.

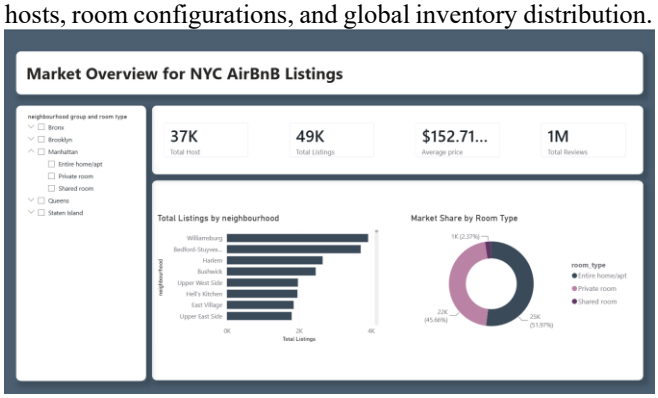
Although the dataset is static, the job can be scheduled to run periodically if new Airbnb data becomes available. This makes the pipeline more reliable, repeatable, and easier to maintain.

F. Dashboard Architecture and Data Modelling

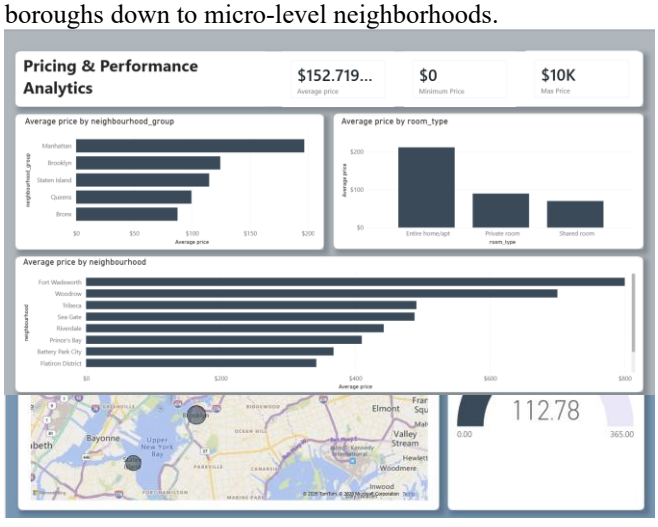
For the data consumption stage, this project uses Power BI as the final serving layer. The cleaned Airbnb dataset was modeled using a simple star schema in the OLAP MySQL database. The central fact table, listingFact, stores listing-level measures such as price, minimum nights, number of reviews, reviews per month, and availability. It is connected to supporting dimension tables such as hostDim and neighborhoodDim, which provide descriptive attributes for hosts and location-based analysis. This model supports analytical queries in Power BI by separating measurable business events from descriptive dimensions.

Although a star schema is recommended for larger analytical systems, this simple star schema is sufficient for this project because the dashboard focuses on listing availability, room type, neighbourhood, price, reviews, and occupancy-related metrics. The Power BI front-end user experience was broken down into three views:

1) Market Overview: A macroscopic view of total active hosts, room configurations, and global inventory distribution.



2) Pricing & Performance Analytics: A comparative breakdown of average nightly rates across macro-level boroughs down to micro-level neighborhoods.



III. Results and Discussion

A. Supply Dynamics and Market Share

The Market Overview dashboard immediately exposes a massive geographical imbalance in NYC's short-term rental ecosystem. Manhattan (21.6K listings) and Brooklyn (20.1K listings) control a staggering 85% of the total market supply. The remaining boroughs (Queens, the Bronx, and Staten Island) function as minor, specialized niche markets.

Furthermore, the data underscores a strong consumer demand for personal space and privacy. Entire homes or apartments make up the majority of the market share at 52.4%, closely followed by private rooms at 45.42%. Shared accommodations, by contrast, represent a negligible slice of the market at just over 2%, proving that communal or shared spaces are largely unpopular in the NYC market, which aligns with broader urban hospitality trends where privacy dictates premium demand [1].

B. Premium Pricing and High-End Micro-Markets

The financial analytics layer confirms that Manhattan commands the highest baseline premium in the city. However, the most actionable visual discovery lies in the exponential price jumps between different property types. Transitioning a property from a simple private room (averaging under $100 USD) to an entire home or apartment ($200+ USD) drives an immediate 136% surge in nightly revenue potential.

Zooming into highly localized neighborhoods reveals deep premium pockets. While Manhattan neighborhoods like Tribeca consistently rank among the most expensive areas due to historical gentrification and location premiums [2], the visualization reveals a surprising outlier: Sea Gate, a gated community in Brooklyn, registers an exceptionally high average nightly rate that competes directly with, and occasionally outpaces, Manhattan's elite districts due to its exclusive, beachfront nature.

C. Spatial Distribution and Occupancy Velocity

Evaluating the Geographical Supply & Occupancy dashboard shows a highly active market, with the citywide estimated occupancy rate sitting at 68.58%. Across the dataset, properties average 114.69 available days per year, pointing to rapid booking cycles and consistent asset turnover.

A deeper look at the map reveals an interesting contrast between Brooklyn and Manhattan. While Manhattan holds the crown for ultra-premium luxury pricing, Brooklyn exhibits the highest density of consistent consumer engagement. Brooklyn listings accumulate a superior volume of total historical reviews, signaling that it serves as the city's primary operational workhorse, drawing in heavy volumes of value-seeking travelers and mid-tier tourists who prioritize affordability and local authenticity [6].

IV. Evaluation and Reflection

The selected tools for this project are MySQL, Pentaho Data Integration, and Power BI. MySQL was used because it provides structured storage and schema validation for the Airbnb dataset. Pentaho was selected because it supports visual ETL development, making it easier to design extraction, cleaning, transformation, and loading steps. Power BI was chosen because it provides interactive dashboard capabilities and supports business-oriented data exploration.

The main advantage of this architecture is that it provides a clear end-to-end data engineering workflow, from raw data ingestion to dashboard consumption. The separation between raw and processed storage also improves reliability and maintainability. However, this design has several limitations. First, the dataset is static, so the pipeline does not fully demonstrate real-time or incremental ingestion. Second, the star schema is relatively simple because the Airbnb dataset has limited relational entities. Third, the solution depends on manual dashboard interpretation, so future work could include automated reporting or machine learning-based price prediction.

V. Conclusion

Data engineering is a vital prerequisite for accurate business intelligence. Without a resilient ETL pipeline, raw public datasets will inevitably corrupt thousands of rows, causing misaligned data fields that skew critical financial metrics like average nightly rates. By implementing data validation, duplicate removal, schema enforcement, and structured loading through Pentaho, this study produced a cleaner and more reliable dataset for Power BI dashboard analysis.

Ultimately, the insights extracted from the Power BI dashboard provide clear strategic direction: Manhattan remains the premier location for high-margin, premium real estate investments, while Brooklyn stands out as a high-occupancy, high-turnover powerhouse that handles the largest share of active guest traffic in New York City.

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